INCH-POUND

MIL-DTL-81984C(OS)
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SUPERSEDING
MIL-C-81984B(AS)
15 May 1989

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, MARK 107 MOD 1

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 107 Mod 1 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-D-21625D	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-D-81303	Design and Evaluation of Cartridges for Stores Suspension Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Team (Code 840M), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 1377

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MIL-I-81577 Ignition Element, Electric, Mark 17 Mod 0

STANDARDS

DEPARTMENT OF DEFENSE

DOD-STD-2101 Classification of Characteristics

(Unless otherwise indicated, copies of Department of Defense specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL613468 Ignition Element, Electric, Mark 17 Mod 0

DL311AS255 Test Fixture, Cartridge, Mark 107 Mod 1

DL745AS100 Cartridge, Impulse, Mark 107 Mod 1

745AS100 Cartridge, Impulse, Mark 107 Mod 1 Assembly

(Application for copies should be addressed to Commander, Naval Surface Warfare Center, Indian Head Division, 101 Strauss Avenue, Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

PUBLICATIONS

OP5 Ammunition and Explosives Ashore

(Application for copies should be addressed to Commander, Naval Ordnance Center, Farragut Hall Bldg. D-323, 23 Strauss Avenue, Indian Head, MD 20640-5555.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

PUBLICATIONS

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Examination

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(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Conformance to documents. The Mark 107 Mod 1 impulse cartridge covered by this specification shall be manufactured in accordance with DL745AS100 and all documents listed thereon.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purposes of this specification the Mark 17 Mod 0 electric ignition element and the cartridge propellant charge are considered primary components.

3.5 Ignition element. The ignition elements shall be furnished from lots which were manufactured within the past12 months. The ignition elements shall be manufactured in accordance with the drawings listed on DL613468, and shall be from lots which were accepted in accordance with the requirements of MIL-I-81577 (see 4.5.1). When the same contractor manufactures the Mark 17 Mod 0 Ignition Element and the Mark 107 Mod 1 Impulse Cartridge and assembles the ignition element in the cartridge, the first article and production test requirements of MIL-I-81577 for the ignition element shall be waived.

3.6 Propellant charge.

3.6.1 Quantity. All propellant charges shall be as specified on drawing 745AS100. Charges shall be determined by weight rather than by volume measurement.

3.6.2 Smokeless powder, handling and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed under conditions that do not permit the relative humidity to exceed 75 percent.

3.6.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.5.2).

3.7 Cartridge inspections.

3.7.1 Bridge circuit resistance. The resistance of the bridge circuit in each completely assembled cartridge, when measured electrically (see 4.5.4) shall be not less than 1.0 ohm nor greater than 1.2 ohms.

3.7.2 Leakage. Cartridge leakage rate shall be not greater than 10^{-5} cubic centimeters of air per second (see 4.5.5).

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3.7.3 X-ray. Cartridges shall be free of imperfections when x-rayed (see 4.5.6).

3.8 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.8.1 through 3.8.3 when subjected to the test prescribed in 4.5.7.

3.8.1 Chamber pressure. The upper (U) and lower (L) limits of maximum pressure in pounds per square inch shall be 7240 and 6470, respectively.

3.8.2 Misfire. There shall be no misfires (see 4.5.7.2).

3.8.3 Hangfire. There shall be no hangfires (see 4.5.7.3).

3.9 Change in propellant. Whenever a change occurs in the propellant lot for the main charge, test firings shall be conducted in accordance with 4.7.

3.10 Workmanship. Cartridges shall be free of the following defects: burrs, dents incomplete or damaged threads, missing or damaged O-ring, perforated closure disc, or other defects which may prevent entry of the cartridge into the firing chamber, or affect the resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper(see 4.5.3).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed as specified in the applicable test method of 4.5.

4.3 First article inspection. First article inspections shall be performed after the award of contract and prior to production by a facility designed in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered as a sample which has met the requirements of 3.2 through 3.6.3 and 3.10 of this specification. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be from the intended sources of supply.

4.3.1 Sample size. The first article sample size shall consist of 94 cartridges.

4.3.2 Inspection routine. Eighty-four cartridges shall be expended in the tests listed below and 10 shall be retained for investigative purposes. Testing shall consist of the following tests as prescribed in table III of MIL-D-81303, except for temperature and humidity cycling (THC) which shall be in accordance with table II of MIL-D-21625D: 40-foot drop test [(6) nonfiring test]; 15 g shock (12); THC (18); vibration (12); −65°F temperature (12); +200°F temperature (12); +70°F temperature(12). The number in parentheses denotes the number of cartridges allocated for each test. After exposure to the specified environmental treatment, the cartridges allocated for the 15 g shock, THC, vibration and normal temperature shall be conditioned for a minimum of 6 hours at 70 ± 5°F and fired within 5 minutes after

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removal from conditioning. The cartridges allocated for the low temperature and high temperature tests shall be conditioned at −65 ± 5°F and 200 ± 5°F, respectively for not less than 6 nor more than 24 hours and fired within 3 minutes after removal from conditioning. All first article cartridge firings shall be conducted in a test fixture in accordance with the requirements of 4.5.7.1. All cartridges shall fire and produce results within the limits specified in 3.8.1 and 3.8.3, except that the upper (U) and lower (L) limits of maximum pressure in pounds per square inch for cartridges fired at −65°F and 200°F shall be U = 7630 and L = 6220. The pressure limits of 3.8.1 apply for all cartridges fired at 70°F.

4.3.3 Failure. Failure of any cartridge shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in table I.

TABLE I. *Quality conformance tests and examinations.*

Item	Nature of test	Requirement para.	Test para.	Classification
Primary component	Single lot verification	3.4	4.4.1	M101
Ignition element	Age and acceptance verification	3.5	4.5.1	C1
Propellant charge	Type and weight	3.6.1	-	M102
Smokeless powder	Condition	3.6.3	4.5.2	M103
Cartridge	Visual examination	3.10	4.5.3	M104
Cartridge	Bridge circuit resistance	3.7.1	4.5.4	C2
Cartridge	Leakage	3.7.2	4.5.5	C3
Cartridge	X-ray	3.7.3	4.5.6	C4
Cartridge	Ballistics	3.8	4.5.7	C5
Cartridge	Ballistics (change in propellant lot)	3.9	4.7	C6

4.4.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 1301 and not greater than 3200. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as critical, major, or minor in accordance with DOD-STD-2102. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

TABLE II. *Quality conformance sample size.*

Lot size	Sample size	Retained sample
66-110	15	10
111-180	15	10
181-300	20	10
301-500	25	10

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501-800	30	10
801-1300	35	10
1301-3200	40	10
3201-8000	50	10
8001-22000	75	10

4.5 Tests.

4.5.1 Ignition element verification. Verification shall be provided that the ignition element used were certified and accepted in accordance with the requirements of 3.5.

4.5.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP5. Evidence of decomposition shall because for rejection.

4.5.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.10 shall be cause for rejection.

4.5.4 Bridge circuit resistance. Conformance with the bridge circuit resistance of 3.7.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to not greater than 25 milliamperes. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 1 percent in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.7.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.5.5 Leakage. Each cartridge in the first article or production sample shall be leak tested in a dry gas leak tester. Cartridges which exhibit a leakage rate greater than 10^{-5} cc per second of air at a pressure differential of 1 ± 0.1 atmosphere shall be considered defective. Cartridges of the first article sample which exhibit a leakage defect shall be cause for rejection. If one or more cartridges of the production lot sample leak, then a leak test of each cartridge purchased under contract for the lot being tested shall be required. If a 100 percent leak test of the lot is required, it shall be at the expense of the contractor. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected (see 6.5).

4.5.6 Radiographic examination. All cartridges in the first article and production test samples shall be radiographically examined in accordance with ASTM E1742. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with the number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. If any assembly or loading defects are noted in the production lot samples, the contractor must x-ray the represented lot 100 percent and submit the x-rays to the cartridge design activity for review. Discontinuities in the serial numbers shall be identified on the x-ray. Defective cartridges found by the vendor shall be clearly identified on the x-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridges with approximately ⅛-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

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4.5.7 Ballistic tests. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for not less than 6 hours at 70 ± 5°F and fired in a closed bomb test fixture manufactured in accordance with the documents listed on DL311AS225. Each cartridge shall be fired within 5 minutes after removal from conditioning. A record of pressure versus time and of the elapsed time between actuation of the firing mechanism and start of pressure rise shall be taken with suitable instrumentation for each firing. The voltage source used for the firing shall 28 ± 2 volts dc, and the firing circuit shall be regulated to produce a firing current of 10 ± 0.5 amperes. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes. Acceptance criteria (regardless of lot size) shall be: accept zero defects and reject if one or more defects.

4.5.7.1 Criteria of acceptability. The upper (U) and lower (L)limits of maximum pressure shall meet the requirements of 3.8.1. A failure of any cartridge to meet this requirement shall be cause for rejection of the lot represented.

4.5.7.2 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.5.7.3 Hangfire. One or more hangfires (see 6.4.1) shall be cause for rejection of the lot represented.

4.6 Retest. There shall be no retest.

4.6.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

4.7 Change in propellant lot. Whenever a change in propellant lot for the main charge occurs, the charge weight shall be verified by firing 36 cartridges which have been temperature conditioned as follows: 12 at −65 ± 5°F, 12 at 70 ± 5°F, and 12 at 200 ± 5°F. The temperature conditioning and firing of these cartridges and the firing results shall be the same as the cartridge allocated for the low, high and normal temperature firing tests specified in 4.3.2.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The Mark 107 Mod 1 impulse cartridge is intended to provide power to stores from the F-14 aircraft. Since the impulse cartridge were designed specifically for use in the F-14 aircraft, there is no commercial application.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2,and 2.3).
c. Whether first article inspection is required, and if so, specify the test activity.
d. Date and place of delivery.
e. Inspection conditions, if other than as specified (see 4.2).
f. That revision "D" of MIL-D-21625 be used (see 4.3.2).
g. Lot size if other than as specified (see 4.4.1).
h. Sampling, if other than as specified (see 4.4.2).
i. That the safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material", DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore", OP5 are applicable.

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of 94 cartridges. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. For the purpose of this specification, the following definitions apply.

6.4.1 Hangfire. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and the start of pressure rise exceeds 0.03 second or time from start of pressure to peak pressure exceeds 0.05 second.

6.5 Advisory leak test. Indian Head leak tests cartridges in accordance with "Cartridge Leak Test Procedures at the Indian Head Division, Naval Surface Warfare Center, Indian Head, MD".

6.6 Subject term (keyword) listing.
F-14
Stores
CAD

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy - OS
(Project 1377-0119)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-81984	2. DOCUMENT DATE (YYMMDD) 000424

3. DOCUMENT TITLE
CARTRIDGE. IMPULSE, MARK 107 MOD 1

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Standardization Team (Code 840M) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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